Sahil Khan

Email | LinkedIn | GitHub | Portfolio | LeetCode

Software engineer focused on backend development and applied AI, with merged open-source contributions.

Projects

AmazonHelp Support Agent **Sep 2026**

Python, Gemini API, sentence-transformers, scikit-learn

- Built intent classification, semantic retrieval, and grounded support-reply generation.
- Added TF-IDF fallback, deterministic support links, and human-escalation rules.
- Implemented human/LLM scoring tools and weighted-kappa agreement analysis.

CodebaseToSpec **Aug 2026**

Python, AST, MongoDB, Pydantic, Gemini API

- Indexed Python files, symbols, and imports in MongoDB using AST parsing.
- Built a bounded tool-calling loop generating structured plans and execution traces.
- Verified file/symbol references and implemented bounded re-planning on errors.

SIS Migration Engine **Jul 2026**

TypeScript, Node.js, GraphQL, PostgreSQL, BullMQ, Redis

- Built staged CSV imports with schema validation, reference checks, and duplicate detection.
- Implemented GraphQL review workflows for editing records and resolving conflicts.
- Queued approved records with BullMQ and isolated failures using PostgreSQL transactions.

StoreIt **Apr 2025 – Jun 2025**

Next.js, TypeScript, Appwrite

- Built file upload and metadata-storage workflows using Next.js and Appwrite.
- Implemented file search, renaming, sharing, and deletion.
- Added failure cleanup to remove uploaded files when metadata creation failed.

CarePulse **Oct 2024 – Nov 2024**

Next.js, TypeScript, Appwrite, Zod, Twilio

- Built patient registration workflows with Zod validation and Appwrite data storage.
- Implemented appointment scheduling and cancellation workflows.
- Integrated Twilio SMS notifications for appointment updates.

Technical skills

Languages: TypeScript, Python, SQL
Backend & Data: Node.js, Express, GraphQL, PostgreSQL, MongoDB, Redis, BullMQ, Knex
AI & Validation: Gemini API, sentence-transformers, scikit-learn, TF-IDF, Pydantic, Zod
Web & Tools: React, Next.js, Git, Docker

Achievements & open source

- Open-source contributor to Vercel's Turborepo: multiple PRs merged to main
- Solved **700+ problems** on LeetCode.

Education

Dr. A.P.J. Abdul Kalam Technical University **Jul 2019 – Jul 2023**
Bachelor of Technology, Computer Science and Engineering *Kanpur, India*

International Indian School **2016 – 2019**
High School (2018–2019) & Secondary School (2016–2017) *Dammam, Saudi Arabia*